

FIN — Release 10.29

Research Programs P323–P336

Exact resource certificates,
refreeze provenance,
and operational transfer boundaries

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Central result. A matching signed-measure/dual-polynomial certificate fixes the continuum seven-moment negative path resource at $\mathcal{N}_{\text{path}} = 0.406706334692258\dots$

Phase provenance. The exact strict phase unit $1/4000$ is the gcd lattice unit generated by the QW-2030 center and QW-2038 grid steps. This is a procedural provenance theorem, not an internal physical source theorem.

Repository. <https://github.com/hyconiek/Fractal-Nadsoliton-Theory>

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Publication statement

This monograph reports all executed Programs P323–P336 and recommends Programs P337–P350. Mathematical proof, high-precision evaluation, target-dependent construction, synthetic design, conditional axiom package, and missing external evidence remain separate. The legacy/strict split, QW-2191, dimensional-source, role-transfer, laboratory, L_{total} , Standard-Model/gravity, and Theory-of-Everything boundaries are preserved.

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Confidence convention

- **[Proven]**: an analytic or finite exact implication is supplied.
- **[Strong evidence]**: deterministic or replicated computation supports the claim inside a declared model.
- **[Moderate evidence]**: the result is robust in a narrower design class.
- **[Speculative]**: a mechanism is proposed without decisive evidence.
- **[Refuted]**: the declared hypothesis class fails.
- **[Blocked by external evidence]**: the result requires a genuine outside record, apparatus, or custody relation.

0.1 Separation of proof layers

Each program distinguishes:

1. a theorem valid for a mathematical class;
2. an evaluation for the frozen FIN numbers;
3. a synthetic design calculation;
4. an external experimental statement.

Numerical agreement cannot move a claim upward in this hierarchy.

0.2 Reproducibility

The deterministic runner uses seed

20260803.

It produces machine-readable JSON, twelve detailed CSV tables, seven figures, and a Lean 4.28 source. External programs P330 and P336 are admission audits: the runner cannot generate their success conditions.

Chapter 1

P323 — Continuum signed-moment duality

1.1 Primal problem

Let

$$h_k = \frac{1}{1 + k^{9/5}}, \quad k = 0, \dots, 6.$$

Among signed Borel measures

$$\mu = \mu_+ - \mu_-$$

on $[0, 1]$ satisfying

$$\int_0^1 x^k d\mu(x) = h_k, \quad k = 0, \dots, 6,$$

minimize the negative Jordan mass

$$\mathcal{N}(\mu) = \mu_-([0, 1]).$$

The shifted moments $h_1, \dots, h_6$ determine a three-node Gaussian rule. Let

$$g(x) = x^3 + c_2x^2 + c_1x + c_0$$

be orthogonal to $1, x, x^2$ for the shifted moment functional. Numerically,

$$\begin{aligned} c_0 &= -0.11092397533402947096\dots, \\ c_1 &= 0.79990303366942818900\dots, \\ c_2 &= -1.63824233917926923454\dots \end{aligned}$$

Its roots are

$$\begin{aligned} r_1 &= 0.23727834187306726370\dots, \\ r_2 &= 0.54819801698629973919\dots, \\ r_3 &= 0.85276598031990223166\dots \end{aligned}$$

The positive weights are

$$\begin{aligned} u_1 &= 0.94183420338052919615\dots, \\ u_2 &= 0.39368551953570941935\dots, \\ u_3 &= 0.07118661177601972793\dots \end{aligned}$$

Define

$$\mu_* = \sum_{j=1}^3 u_j \delta_{r_j} - \mathcal{N}_* \delta_0, \quad \mathcal{N}_* = \sum_j u_j - 1.$$

Then

$$\mathcal{N}_* = 0.40670633469225834343\dots$$

and all seven moment residuals are below 3.1×10^{-92} in the 90-digit computation.

1.2 Dual problem

For any polynomial

$$-1 \leq p(x) \leq 0 \quad \text{on} \quad [0, 1],$$

and any feasible signed measure,

$$\sum_{k=0}^6 p_k h_k = \int p d\mu \leq \mu_-([0, 1]).$$

This is weak duality.

Set

$$q(x) = \frac{g(x)}{g(0)}, \quad p_*(x) = -q(x)^2.$$

The polynomial p_* vanishes at all three positive atoms and equals -1 at the negative atom $x = 0$. Its coefficients in ascending order are

$$\begin{aligned} &(-1, 14.4225453741745, -81.5405646086897, 231.037739077899, \\ &-348.146892117430, 266.291491557255, -81.2735348088558). \end{aligned}$$

The two internal critical points of q have squared magnitudes approximately 0.01060 and 0.00987, while $q(1)^2 \approx 0.20922$. Thus $q(x)^2 \leq 1$ on $[0, 1]$, so p_* is dual-feasible.

Finally,

$$\sum_{k=0}^6 (p_*)_k h_k = 0.40670633469225834343\dots = \mathcal{N}_*.$$

Matching primal and dual values prove optimality.

1.3 Interpretation

The strict attenuation cannot be generated by a positive distribution of exponential distance scales. Within this seven-moment representation, the minimum signed resource is approximately 0.4067063, not merely nonzero.

This does **not** prove:

- a quantum quasiprobability;
- negative thermodynamic entropy;
- physical information destruction;
- an open-system reservoir;
- a unique microscopic path model.

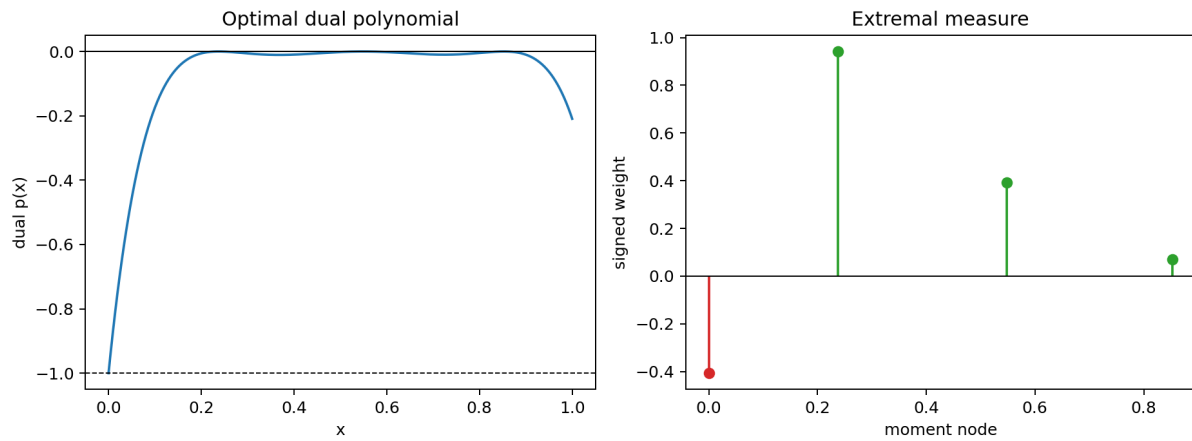


Figure 1.1: The matching continuum signed-measure and dual-polynomial extremizers.

Chapter 2

P324–P325 — Source independence and phase provenance

2.1 P324: completion fibers

Let A_L be the repaired positive legacy operator. Consider a deterministic completion rule

$$F : A_L \mapsto A_T.$$

Three distinct positive radial target operators were constructed on the same carrier and with exactly the same input A_L :

1. the frozen strict target;
2. an alternative with $(\omega, \phi, \beta, \eta) = (0.20175, 0.11250, 0.92, 1.7)$;
3. an alternative with $(0.16975, 0.06250, 0.84, 1.6)$.

For each target A_T , the block operator

$$\mathcal{P}_T = \begin{pmatrix} A_L & A_L^{1/2} A_T^{1/2} \\ A_T^{1/2} A_L^{1/2} & A_T \end{pmatrix}$$

is positive up to numerical roundoff and has cross support. But the cross block explicitly reads A_T . Therefore:

$$A_{T,1} \neq A_{T,2} \implies \neg(F(A_L) = A_{T,1} \wedge F(A_L) = A_{T,2}).$$

The common-parent construction is a representation theorem, not a source-independent generation theorem.

The newly defined object O76 is the **completion fiber**

$$\mathfrak{F}(A_L) = \{A_T : A_T \text{ satisfies the declared target-side admissibility rules}\}.$$

A source law must select one point of this fiber without reading the chosen target in advance.

2.2 P325: exact lattice genealogy

The QW-2030 upstream center was

$$\omega_0 = \frac{187}{800}, \quad \phi_0 = -\frac{11}{80}.$$

The QW-2038 grids used exact decimal steps

$$\Delta\omega = 0.016 = \frac{2}{125}, \quad \Delta\phi = 0.05 = \frac{1}{20}.$$

The selected row is obtained by the grid indices $(-3, +6)$:

$$\omega_* = \omega_0 - 3\Delta\omega = \frac{743}{4000},$$

$$\phi_* = \phi_0 + 6\Delta\phi = \frac{13}{80} = \frac{650}{4000}.$$

The additive subgroup of $\mathbb{Q}$ generated by the two centers and two steps is

$$\left\langle \frac{187}{800}, \frac{2}{125}, -\frac{11}{80}, \frac{1}{20} \right\rangle_{\mathbb{Z}} = \frac{1}{4000}\mathbb{Z}.$$

Hence the generator

$$r = e^{i/4000}$$

is minimal for the **procedural refreeze lattice**.

2.3 Falsification of an internal-source reading

The provenance chain uses:

- supplied upstream center values;
- investigator-chosen grid widths and step counts;
- a mass/flavor/GW objective;
- threshold gates;
- selection of the best passing row.

This is a real derivation of the selected grid coordinate from a declared search protocol. It is not a derivation of the coordinate from the informational nadsoliton alone.

Accordingly:

Claim	Verdict
$e^{i/4000}$ exactly generates the frozen decimal phase tuple	Proven
$1/4000$ is the ged unit of the QW-2030/QW-2038 rational lattice	Proven
the lattice unit is target-independent of the refreeze procedure	Refuted
the lattice unit is a strict ontological source	Not established
the lattice unit discharges QW-2191	Refuted

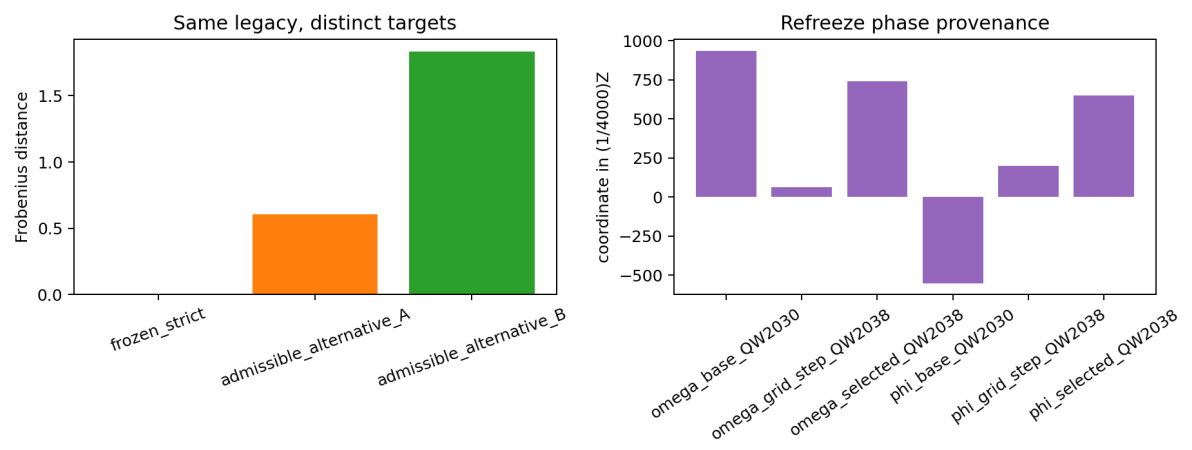


Figure 2.1: One legacy input with distinct target parents, and the exact QW-2030/QW-2038 phase-lattice genealogy.

Chapter 3

P326–P331 — Operational and transfer programs

3.1 P326: loss-robust finite experiment

The experiment class uses:

- all twelve vertex preparations;
- a twelve-outcome vertex detector plus a no-click outcome;
- mean efficiency 0.82 with synthetic component spread 0.025;
- nearest-neighbor confusion near 0.02;
- relative clock jitter with standard deviation 0.004;
- 48 nuisance draws;
- scale-aligned legacy dynamics.

For heat dynamics the best robust row is

$$t = 0.73863, \quad x = 8, \quad \text{TV}_{0.05} = 0.18579,$$

with conservative shot bound 214.

For wave dynamics:

$$t = 1.64532, \quad x = 8, \quad \text{TV}_{0.05} = 0.34147,$$

with conservative shot bound 64.

These are fifth-percentile detector-level distances under the supplied nuisance family. They show that the separation found in P315 is not erased by this particular loss/confusion model.

They do not establish the nuisance family of a real detector.

3.2 P327: exact chamber atlas

For positive radial weights $w_1, \dots, w_6$ on C_{12} , the six nonzero mode values are linear:

$$\lambda = w^T M,$$

where

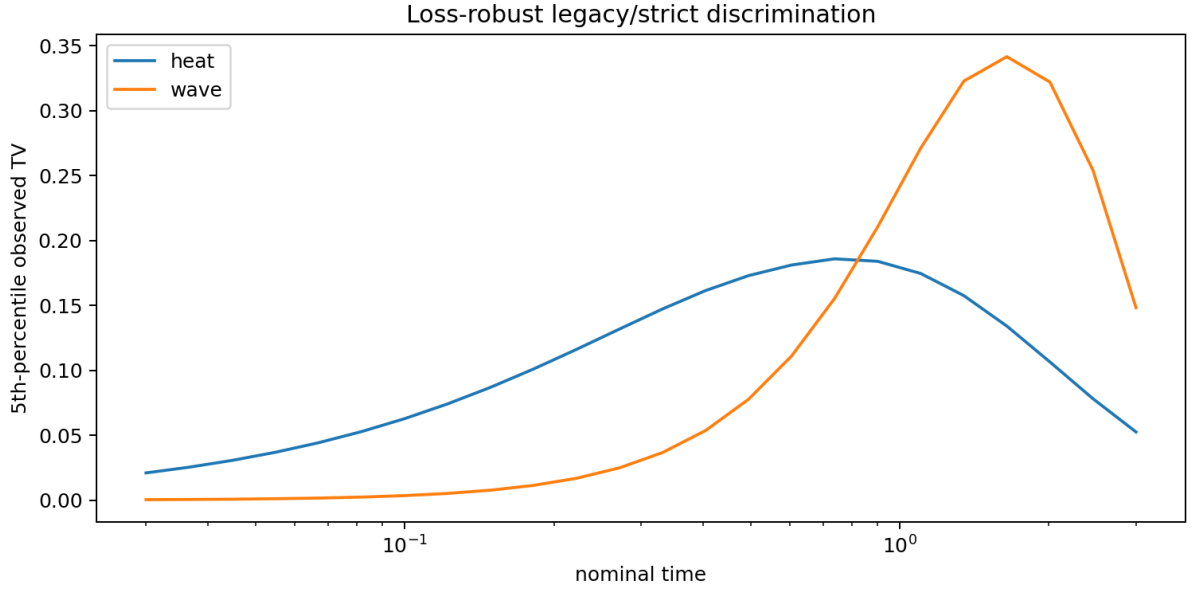


Figure 3.1: Detector-degraded fifth-percentile operational separation under the declared synthetic nuisance model.

$$M = \begin{pmatrix} 2 - \sqrt{3} & 1 & 2 & 3 & 2 + \sqrt{3} & 4 \\ 1 & 3 & 4 & 3 & 1 & 0 \\ 2 & 4 & 2 & 0 & 2 & 4 \\ 3 & 3 & 0 & 3 & 3 & 0 \\ 2 + \sqrt{3} & 1 & 2 & 3 & 2 - \sqrt{3} & 4 \\ 2 & 0 & 2 & 0 & 2 & 0 \end{pmatrix},$$

and

$$\det M = -3456\sqrt{3} \neq 0.$$

At the uniform point $w_0 = (1/6, \dots, 1/6)$, all mode values equal 2. Let q be any permutation of

$$(-5, -3, -1, 1, 3, 5).$$

Define

$$c(q) = -\frac{q_1 + q_3 + q_5}{3}$$

and

$$w(q) = \frac{1}{6}\mathbf{1} + \frac{1}{100}M^{-T}(q + c(q)\mathbf{1}).$$

Then

$$\sum_i w_i(q) = 1,$$

and the corresponding mode values are

$$\lambda_i(q) = 2 + \frac{1}{100}(q_i + c(q)).$$

Exact enumeration over all 720 permutations proves:

$$\min_{q,i} w_i(q) = \frac{7}{50} > 0,$$

and every adjacent mode gap is at least $1/50$.

Thus every one of the 720 strict orders occurs in the interior of the positive radial simplex. Spectral order cannot be a selector.

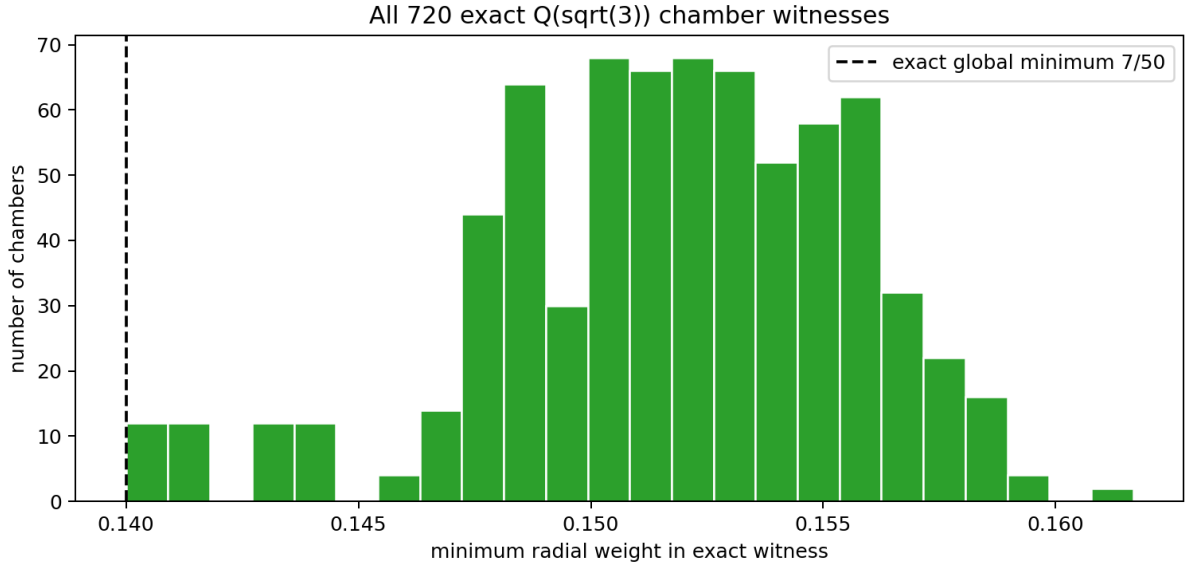


Figure 3.2: Minimum positive weights for all 720 exact algebraic spectral-order witnesses.

3.3 P328: one-time wave-channel diamond distance

For unitary channels

$$\Phi_U(\rho) = U\rho U^\dagger, \quad \Phi_V(\rho) = V\rho V^\dagger,$$

let ν be the distance from the origin to the convex hull of the eigenvalues of $U^\dagger V$. Then

$$\frac{1}{2}\|\Phi_U - \Phi_V\|_\diamond = \sqrt{1 - \nu^2}.$$

Because the audited legacy and strict wave generators are circulant, their relative phases are simultaneously diagonalizable. On the scanned time range the half diamond distance reaches

$$1,$$

with the first sampled perfect-separation time

$$t \approx 0.75591.$$

This means that an optimal single-use quantum input and measurement can perfectly distinguish the two ideal unitary channels at that sampled time.

It does not mean that the present photonic vertex protocol realizes the optimal ancilla-assisted strategy. A true multitime comb additionally requires interventions, memory spaces, and an SDP over declared instruments.

3.4 P329: quartic-clock minimax design

The clock family is

$$\tau(t) = t + a_2 t^2 + a_3 t^3 + a_4 t^4,$$

with

$$|a_2| \leq 0.08, \quad |a_3| \leq 0.04, \quad |a_4| \leq 0.02, \quad \tau'(0) = 1.$$

The external slope calibration removes the exact first-order degeneracy between generator scale and a free linear clock coefficient.

Among 4368 five-time designs, evaluated on 27 corner/interior clock tuples, the best design is

$$(0.07319, 0.09360, 0.52349, 1.09477, 1.40000).$$

Its worst-case minimum Fisher eigenvalue is

$$7.2765 \times 10^{-4},$$

which is 3.86 times the reference design value.

The result is minimax only for the declared quartic family and coefficient box. Arbitrary monotone or nonparametric time warps remain capable of reopening nonidentifiability.

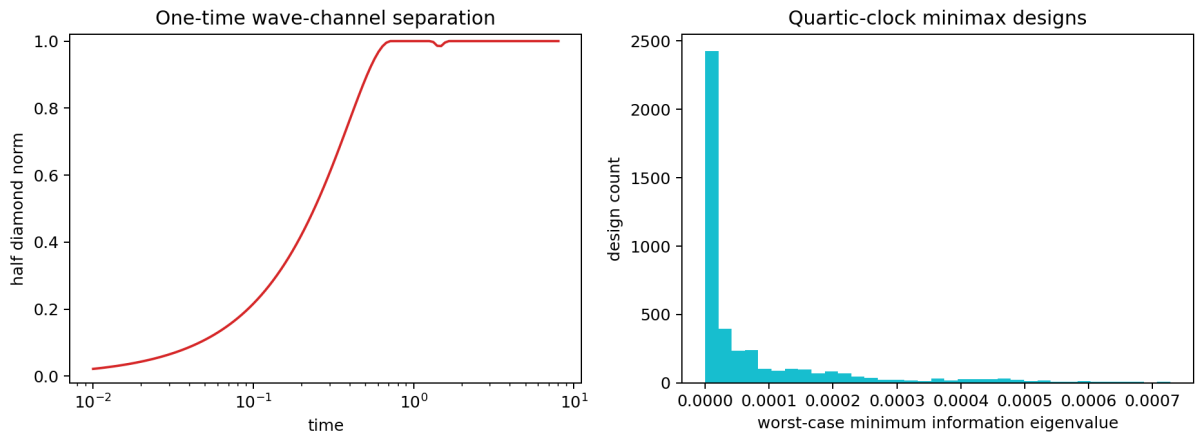


Figure 3.3: Ideal one-time wave-channel diamond separation and the quartic-clock minimax design distribution.

3.5 P330: laboratory admission

No manifest satisfying all of the following was admitted:

1. distinct provider, registrar, and analyst;
2. frozen hold-out hash;
3. raw event-file hash;
4. apparatus calibration;
5. non-template production status.

Validator 241 and Pipeline 242 were therefore not run.

3.6 P331: photonic transfer specification

The best wave witness requires a dense signed 12-mode coupling with nonzero coupling dynamic range approximately

$$42.45.$$

For witness time $t = 1.64532$, a conservative Duhamel allocation of output TV budget 0.05 gives generator-norm budget

$$\|\Delta A\| \lesssim \frac{0.05}{t} = 0.03039.$$

The synthetic tolerance sweep shows:

Relative coupling error SD	p95 output TV error	Pass 0.05 budget
10^{-4}	1.25×10^{-4}	yes
10^{-3}	1.16×10^{-3}	yes
10^{-2}	1.19×10^{-2}	yes
3×10^{-2}	3.60×10^{-2}	yes
5×10^{-2}	5.99×10^{-2}	no
10^{-1}	1.24×10^{-1}	no

The proposed implementation hypothesis is:

1. heralded single-photon source;
2. programmable twelve-mode interferometer;
3. phase and amplitude controls;
4. twelve SPAD/SNSPD channels with TCSPC;
5. independent clock and registrar;
6. raw event-level record.

Because the coupling is dense and signed, a programmable interferometer or unitary dilation is more plausible than a passive nearest-neighbor waveguide array.

The pre-unblinding failure rule is mandatory: if the frozen transfer-matrix or detector hold-out exceeds its budget, reject the implementation without refitting FIN.

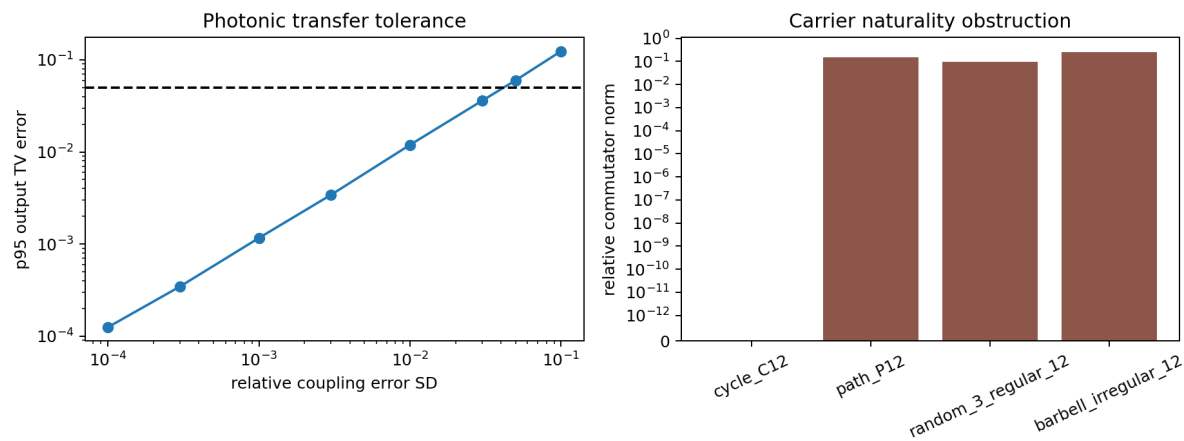


Figure 3.4: Photonic transfer tolerance and noncyclic carrier commutator obstructions.

Chapter 4

P332–P336 — Interpretation, carriers, axioms, and independence

4.1 P332: signed-resource ambiguity

The P323 signed measure can be written

$$\sum_j u_j \delta_{r_j} - \mathcal{N}_* \delta_0.$$

At the level of scalar moments, the negative term may equally be written as

$$\mathcal{N}_* e^{i\pi} \delta_0.$$

Thus a real signed Jordan representation and a phase- π complex-amplitude representation give the same scalar sequence.

Other interpretations require additional structure:

Interpretation	Additional structure	Not supplied by moments
quasiprobability	state/effect frame and free operations	operational negativity monotone
complex interference	phase reference and Born map	detector probability law
Krein space	fundamental symmetry and positive physical subspace	probability interpretation
open system	environment and CPTP dilation	process tensor
active feedback	controller, energy ledger, causal interventions	work and control record

Therefore the signed path resource is mathematically unavoidable in its declared representation, while its physical meaning is underdetermined.

4.2 P333: carrier naturality

If

$$A_S = f(A_L)$$

by scalar functional calculus, then

$$[A_L, A_S] = 0.$$

The relative commutator audit gives:

Carrier	Relative commutator
cycle C_{12}	5.2×10^{-17}
path P_{12}	0.1482
random 3-regular graph	0.09356
irregular barbell graph	0.26046

Thus scalar functional completion is not carrier-natural across the tested noncyclic family. The cycle is special because its radial operators share a Fourier basis.

This does not exclude:

- nonscalar completion;
- carrier-dependent completion;
- an enlarged Hilbert space;
- a target-importing joint parent.

4.3 P334: minimal conditional electroweak package

Take the historical relation only as an explicit axiom:

$$x = \sin^2 \theta_W = \frac{\alpha_{\text{geo}}}{12} = 0.2310490602 \dots$$

Once the $U(1)$ and $SU(2)$ sectors are ordered, the coupling ratio is

$$\frac{g'}{g} = \sqrt{\frac{x}{1-x}} = 0.5481542518 \dots$$

The sector swap gives

$$x \mapsto 1 - x, \quad \frac{g'}{g} \mapsto \frac{g}{g'},$$

so an unpointed sector pair cannot select the observed branch.

The minimal angle-only package contains:

1. one discrete sector-pointing axiom;
2. one role-law axiom $x = \alpha_{\text{geo}}/12$;
3. zero new continuous parameters for the angle alone.

A minimal tree-level mass scaffold additionally needs at least:

1. an overall gauge coupling g ;
2. a symmetry-breaking scale v ;
3. representation and covariant-derivative laws;
4. a symmetry-breaking sector;
5. a mass/observable map.

No full electroweak theory follows from the current spectrum.

4.4 P335: independence cube

Let the resource vector be

$$R = (N, \Phi, C, P, S) \in \{0, 1\}^5,$$

where:

- N : signed path resource;
- Φ : nontorsion phase;
- C : independent cross law;
- P : pointing;
- S : dimensional scale.

All $2^5 = 32$ product-signature models were constructed. For each resource, there exists a model in which the other four are present and that resource is absent:

$$01111, \quad 10111, \quad 11011, \quad 11101, \quad 11110.$$

Therefore no implication

$$R_i \Leftarrow \bigwedge_{j \neq i} R_j$$

is valid in the minimal uncoupled signature.

This result is formalized in Lean 4.28. It proves logical independence before coupling laws are added. It does not prove that a future FIN dynamical law cannot co-generate several resources.

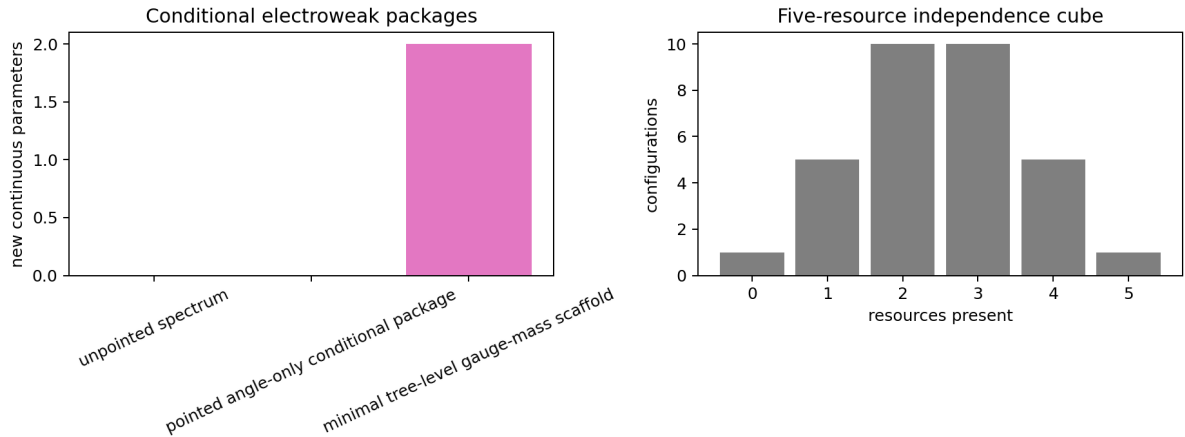


Figure 4.1: Conditional electroweak parameter accounting and the five-resource independence cube.

4.5 P336: reservoir gate

No package contains certified:

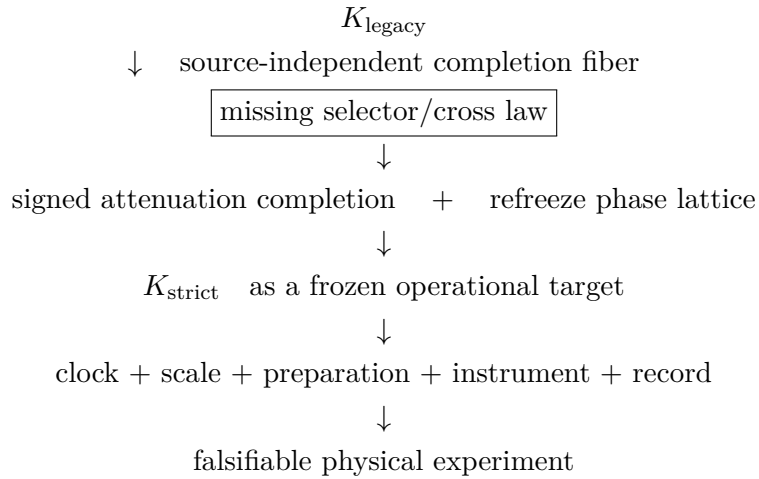
- physical hardware data;
- paired controls and matched seeds;
- frozen primary endpoint;
- calibration record;
- independent custody.

The external reservoir experiment remains blocked.

Chapter 5

Cross-program synthesis

5.1 The refined bridge-resource diagram



The diagram contains two different notions of "generation":

1. mathematical construction after target data are supplied;
2. source-independent derivation before the target is known.

Current FIN has strong examples of the first and open obligations for the second.

5.2 A smaller object does not yet replace the resource tuple

P335 shows that no one resource follows logically from the other four in the minimal signature. Consequently, replacing the five-component budget by one unnamed "deeper object" would currently hide assumptions rather than reduce them.

A legitimate smaller object must arrive with coupling theorems proving that its projections generate:

$$(N, \Phi, C, P, S).$$

No such theorem is currently exported.

5.3 What the phase result means

P325 changes the interpretation of O66 from Release 10.28.

The phase generator $e^{i/4000}$:

- is exactly minimal for the frozen rational tuple;
- is exactly minimal for the QW-2030/QW-2038 scan lattice;
- is not contained in the finite legacy phase group;
- is procedurally generated by the refreeze design;
- is not yet internally generated by strict informational dynamics.

This is a sharper and more informative result than calling $1/4000$ an arbitrary fitted decimal. It also prevents promoting a procedural grid unit into an ontological constant.

5.4 Operational consequences

Three different distance levels are now available:

1. vertex-measurement TV distance;
2. detector-degraded robust TV distance;
3. ideal one-time quantum-channel diamond distance.

They answer different questions and must not be conflated. In particular, the ideal diamond distance reaching one does not eliminate the need to compile and calibrate the corresponding optimal input and measurement.

Chapter 6

Failed approaches and no-go boundaries

Candidate	Verdict	Reason
the 0.4067 negativity was only a grid artifact	Refuted	matching continuum primal–dual certificate
1/4000 is presently an internal FIN constant	Refuted on current provenance	it is the gcd unit of supplied scan centers and steps
a common parent generates strict automatically	Refuted	cross block reads the chosen target
spectral order selects strict	Refuted exactly	all 720 orders have positive algebraic witnesses
detector loss erases all legacy/strict difference	Refuted in supplied nuisance class	robust TV remains 0.186/0.341
one-time ideal wave channels are operationally equivalent	Refuted	half diamond norm reaches one
clock design alone fixes an uncalibrated linear clock	Refuted	scale and clock-slope columns coincide
signed mass uniquely means quantum interference	Refuted	several inequivalent realizations share the moments
cyclic scalar completion is carrier-universal	Refuted in tested family	three noncyclic commutators are nonzero
unpointed spectrum gives the electroweak branch	Refuted	sector swap sends x to $1 - x$
four bridge resources imply the fifth	Refuted in minimal signature	five single-omission models
repository output supplies external experiment	Blocked	custody, calibration, raw data, hold-out absent

Chapter 7

Recommended Programs P337–P350

Probabilities estimate the chance of a decisive result, including a rigorous negative result.

Rank	Program	Study	Decisive output	Probability
1	P337	formal algebraic/interval P323 certificate	machine-check node, weight, and dual-feasibility enclosures	0.91
2	P338	full oscillatory-kernel signed resource	extend the moment extremizer from attention to the complete signed strict kernel	0.84
3	P339	continuous refreeze sensitivity audit	distinguish stable objective basin from grid-coordinate artifact on frozen hold-out	0.88
4	P340	natural common-parent theorem	classify carrier-natural transformations across cycles, paths, and irregular graphs	0.82
5	P341	compiled photonic inverse calibration	convert the P331 matrix budget into a mesh and calibration-inversion executable	0.83
6	P342	independent P241 production run	admit, validate, hash, and execute one real event bundle	0.28
7	P343	multitime quantum-comb SDP	compute an adaptive operational distance for declared interventions	0.76
8	P344	nonparametric clock bounds	replace the quartic box by monotone Sobolev or spline uncertainty	0.79
9	P345	carrier-universal strict invariants	identify which phase/compression features survive graph changes naturally	0.75
10	P346	bridge-resource monotones	define free completion maps and monotones for negativity, phase, pointing, and scale	0.74
11	P347	resource coupling theorem search	test whether one strict law can co-generate two or more Q86 factors	0.68
12	P348	falsifiable conditional EW package	preregister observables that are not algebraic restatements of the angle axiom	0.72
13	P349	dimensional conversion metrology	specify operational calibration of $\hbar_*, \ell_*, \tau_*$ without calling them emergent	0.70
14	P350	external reservoir execution	run paired hardware and control records under a frozen endpoint	0.25

7.1 Recommended immediate sequence

$$P337 \longrightarrow P338 \longrightarrow P339 \longrightarrow P340 \longrightarrow P341 \longrightarrow P343$$

The sequence first upgrades the strongest new mathematical result, then includes the oscillatory phase, audits the actual selection procedure, demands carrier naturality, compiles a physical transfer design, and finally constructs the correct multitime operational distance.

P342 and P350 remain external and cannot be completed by repository simulation.

Chapter 8

Research roadmap

8.1 Mathematical track

The highest-value mathematical tasks are:

1. formalize the P323 extremal certificate;
2. determine the signed resource of the full kernel, not only its envelope;
3. classify natural parent transformations;
4. define resource monotones and coupling laws.

Success may be either a constructive theorem or a no-go theorem.

8.2 Operational track

The operational track should:

1. compile the strict unitary into a declared optical architecture;
2. invert calibration records into a confidence set for the implemented channel;
3. freeze preparation, time, measurement, and failure thresholds;
4. keep provider, registrar, and analyst separate;
5. report a failed calibration or null result without model repair.

8.3 Conditional-physics track

Conditional packages may be developed, but each supplied object must remain visible:

informational core + conversion axioms + sector axioms + apparatus model.

No conditional theorem is to be renamed a strict zero-parameter prediction.

Chapter 9

Limitations

1. The P323 theorem is analytic, while the displayed FIN evaluation is a 90-digit computation rather than a fully checked algebraic-number proof.
2. P324 alternatives demonstrate underdetermination; they are not claimed as physical competitors.
3. P325 proves procedural phase provenance, not causal physical origin.
4. P326, P329, and P331 use synthetic nuisance families.
5. P327 classifies order only, not spectral magnitude or physical likelihood.
6. P328 treats ideal one-use unitary wave channels, not a full comb.
7. P333 tests four carriers and one necessary scalar-calculus condition.
8. P334 is explicitly axiom-augmented.
9. P335 proves independence only before additional coupling laws.
10. P330 and P336 remain externally blocked.

Chapter 10

Final conclusion

The deepest result of this round is not that another physical role has been found. It is that three previously blurred issues have been separated exactly.

The strict attenuation has a quantitatively exact signed-moment resource:

$$\mathcal{N}_{\text{path}}^{\min} = 0.406706334692258 \dots$$

The frozen strict phase has an exact procedural lattice source:

$$\frac{1}{4000}\mathbb{Z},$$

generated by the QW-2030 center and QW-2038 scan increments.

And the remaining bridge resources are not consequences of one another in the minimal uncoupled signature.

Therefore the current scientifically surviving interpretation is:

FIN contains a mathematically coherent informational operator target with exact spectral, moment, phase-lattice, and operational certificates. Its legacy-to-strict completion and its transition to physics require separately declared source, sector, scale, instrument, and record structures. Some of those structures can be constructed conditionally; none has yet been shown to emerge as one unavoidable strict physical law.

This is stronger than an interpolation claim and weaker than a physical theory. It is the correct boundary for the next research round.

Chapter 11

References

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10. Existing FIN repository packets K1, K2, F2, F3, S2, QW-2030, QW-2038, QW-2049, P295–P322, and the Release 10.28 executable artifacts.

Reproduction certificate

```
MPLCONFIGDIR=/tmp/mpl-fin \  
python3 -m unittest -v test_fin_programs_323_336.py
```

Expected result: 18 tests passed, 0 failed. The suite recompiles the dependency-free Lean 4.28 source. Detector, clock, coupling-error, and finite-count assumptions are synthetic. P330 and P336 admit no external laboratory or hardware evidence.

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